Assignment -3

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1. Create a 'Question' class in a quiz application (Hint: There is a quiz application, where it contains loads of MCQs (Multiple Choice Questions). Each question would have two, three or four options - like option A, B, C, D.)

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Assignment3
{
    internal class Question
    {
        public string Text;
        public List<string> Options;
        public char CorrectOption;

        public Question(string text, List<string> options, char correctOption)
        {
            Text = text;
            Options = options;
            CorrectOption = correctOption;
        }

        public void Display()
        {
            Console.WriteLine(Text);
            char label = 'A';
            foreach (string option in Options)
            {
                Console.WriteLine($"{label}. {option}");
                label++;
            }

        }

        public bool CheckAnswer(char userAnswer)
        {
            return char.ToUpper(userAnswer) == CorrectOption;
        }
    }
}
namespace Assignment3
{
internal class Program
{
        static void Main(string[] args)
        {
                Question q1 = new Question(
        "What is the capital of France?",
        new List<string> { "Berlin", "Madrid", "Paris", "Rome" },
        'C'
);
                q1.Display();
                char userAnswer = 'C';    // pretend the user typed this
                bool isCorrect = q1.CheckAnswer(userAnswer);

                Console.WriteLine(isCorrect ? "Correct!" : "Wrong answer.");
        }
}
}
```

```
What is the capital of France?
A. Berlin
B. Madrid
C. Paris
D. Rome
Correct!

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Press any key to close this window . . .
```

2. Create a C# console application that demonstrates interfaces in an app for college / Univeristy 'Library' environment.

```csharp
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Assignment3
{
    internal interface ILibraryItem
    {
        string Title { get; set; }
        void Borrow();
        void Return();
        void DisplayInfo();
    }
}


using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Assignment3
{
    internal class Book : ILibraryItem
    {
        public string Title { get; set; }
        public string Author;
        private bool isBorrowed = false;

        public Book(string title, string author)
        {
            Title = title; Author = author;
        }

        public void Borrow()
        {
            if (isBorrowed)
            {
                Console.WriteLine($"'{Title}' is already borrowed.");
            }
            else
            {
                isBorrowed = true;
                Console.WriteLine($"You borrowed '{Title}' by {Author}.");
            }
        }

        public void Return()
        {
            isBorrowed = false;
            Console.WriteLine($"You returned '{Title}'.");
        }

        public void DisplayInfo()
```

```
        {
            Console.WriteLine($"Book: {Title} by {Author} | Borrowed: {isBorrowed}");
        }
    }
}
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Assignment3
{
    internal class DVD : ILibraryItem
    {
        public string Title { get; set; }
        public int DurationMinutes;
        private bool isBorrowed = false;

        public DVD(string title, int durationMinutes)
        {
            Title = title;
            DurationMinutes = durationMinutes;
        }

        public void Borrow()
        {
            if (isBorrowed)
            {
                Console.WriteLine($"'{Title}' DVD is already checked out.");
            }
            else
            {
                isBorrowed = true;
                Console.WriteLine($"You checked out the DVD '{Title}' ({DurationMinutes}
min).");
            }
        }

        public void Return()
        {
            isBorrowed = false;
            Console.WriteLine($"You returned the DVD '{Title}'.");
        }

        public void DisplayInfo()
        {
            Console.WriteLine($"DVD: {Title} ({DurationMinutes} min) | Borrowed:
{isBorrowed}");
        }
    }
}
static void Main(string[] args)
{
      List<ILibraryItem> libraryItems = new List<ILibraryItem>
{
new Book("Clean Code", "Robert C. Martin"),
new DVD("Inception", 148)
};

      foreach (ILibraryItem item in libraryItems)
      {
              item.DisplayInfo();
              item.Borrow();
              item.Borrow();   // try borrowing again to see the "already borrowed"
              message
              Console.WriteLine();
      }
    }
```

```
Book: Clean Code by Robert C. Martin | Borrowed: False
You borrowed 'Clean Code' by Robert C. Martin.
'Clean Code' is already borrowed.

DVD: Inception (148 min) | Borrowed: False
You checked out the DVD 'Inception' (148 min).
'Inception' DVD is already checked out.


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Press any key to close this window . . .
```

3. Create a C# program that defines a class containing a property named Pin, and implement validation logic within the property to ensure only valid PIN values are accepted.

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Assignment3
{
    internal class BankCard
    {
        private int pin;   // the REAL storage – kept private, hidden from outside

        public int Pin
        {
            get
            {
                return pin;
            }
            set
            {
                if (value >= 1000 && value <= 9999)
                {
                    pin = value;
                    Console.WriteLine($"PIN set successfully to {value}.");
                }
                else
                {
                    Console.WriteLine($"Invalid PIN '{value}' – PIN must be a 4-digit
number between 1000 and 9999. Keeping previous value.");
                }
            }
        }
    }
}


static void Main(string[] args)
{
    BankCard card = new BankCard();

    card.Pin = 4521;      // valid
    card.Pin = 42;         // invalid – too short
    card.Pin = 99999;      // invalid – too long

    Console.WriteLine("Final stored PIN: " + card.Pin);
}
```

```
PIN set successfully to 4521.
Invalid PIN '42' – PIN must be a 4-digit number between 1000 and 9999. Keeping previous value.
Invalid PIN '99999' – PIN must be a 4-digit number between 1000 and 9999. Keeping previous value.
Final stored PIN: 4521

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```